Bioinformatics Learning Order

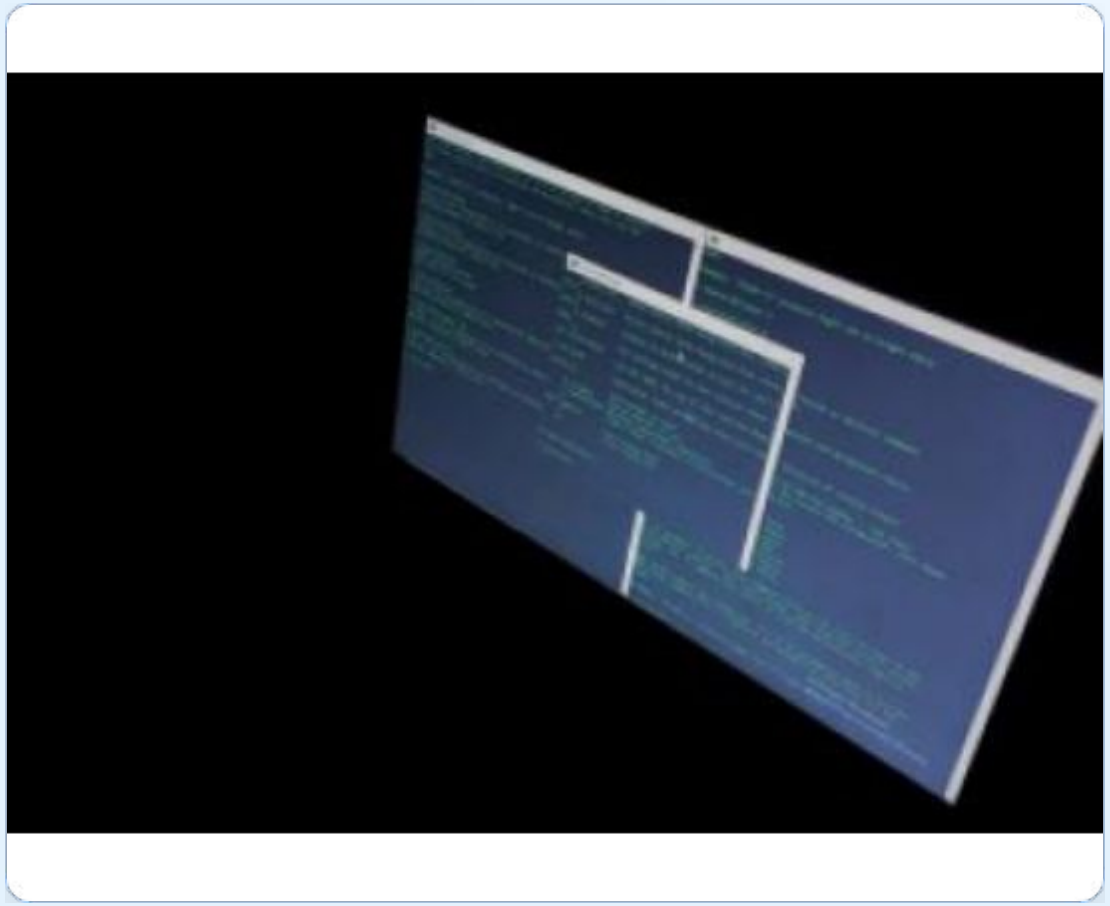
A pattern for success in computational biology

buluthamali.com

Linux before HPC

Master the CLI First

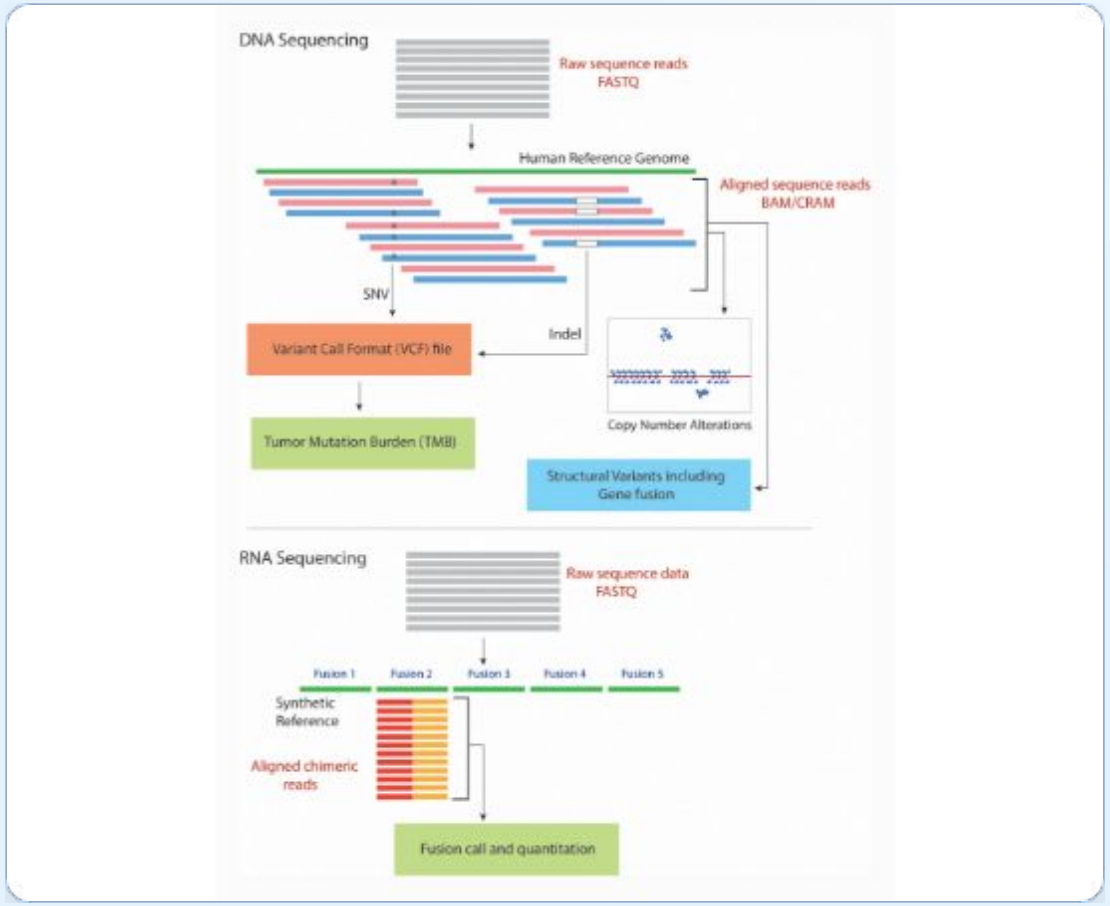
Understanding the Linux environment is highly valuable before scaling to High Performance Computing (HPC). You should be able to navigate, manage files, and understand permissions before managing cluster jobs.



Bash before Workflow Systems

Scripts to Pipelines

Before using Nextflow or Snakemake, it helps to understand how to chain commands in Bash. Modern workflow managers are powerful, but they build on fundamental shell scripting concepts.



Statistics before AI/ML

Foundations of AI

Artificial Intelligence and Machine Learning build on core statistical modeling ideas. Understanding distributions, P-values, and regression is essential for meaningful and trustworthy ML applications in biology.

QC before Interpretation

Quality Control

Verify data integrity, check for contamination, and assess sequencing depth.

Transition

Garbage in, garbage out; clean data leads to more robust findings.

Interpretation

Extract biological insights only after you are confident in the underlying data quality.

FASTQ before RNA-seq

Understand Raw Data

Before running differential expression, you should understand the FASTQ format and the information contained in Phred scores. This raw layer underpins every downstream RNA-seq step.

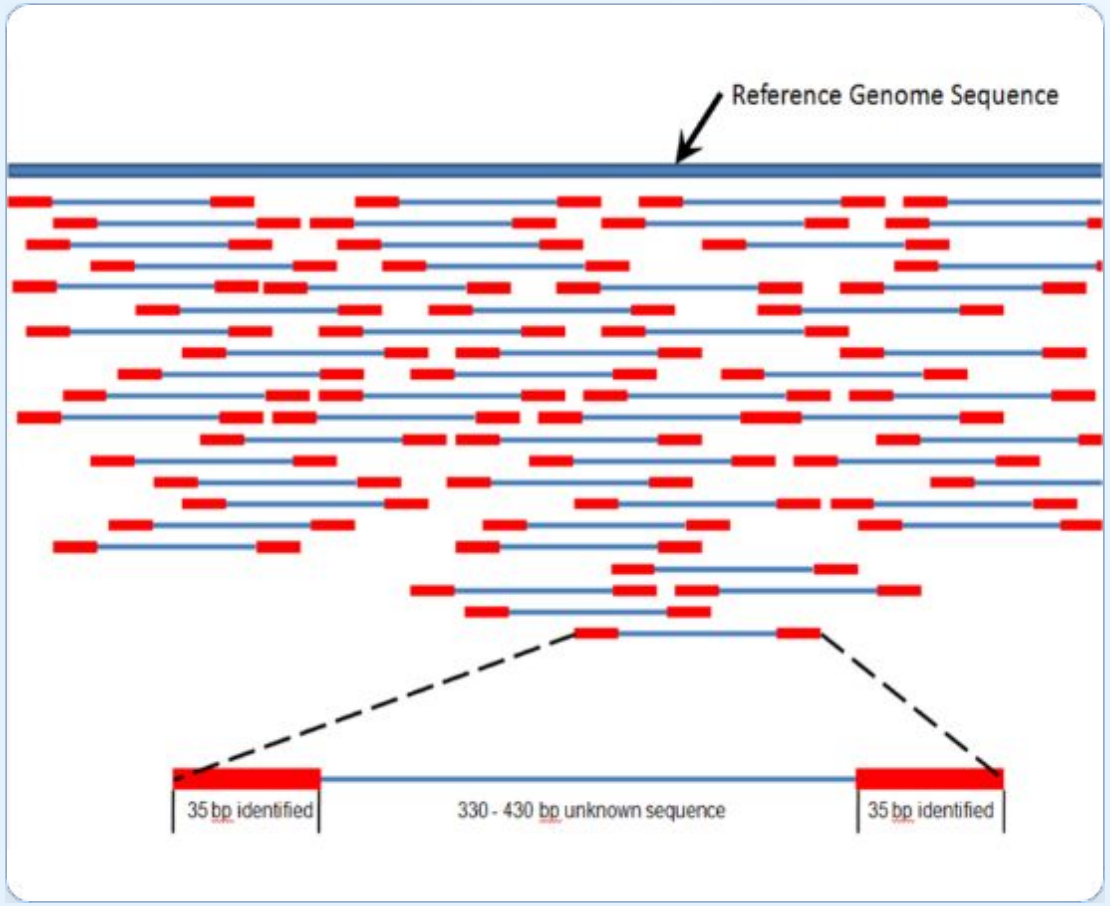
```
@HWI-ST911:111:C0N4WACXX:5:1101:2249:2216 1:N:0:TTAGGC CGATC:@@@FF
NATGGCACCATTAAAAAGAATGTTTTATATGGTGTTGAGAAGGACAAAGCTGAAGAAGAAATTTAGTCTGCACTTGATGTTGCAAATGCAAAGAAA
+
#2A2<CCFHIIIIIIIIIIIIGCCHIIIGIIIFFHIIDGHIGIIIIIIICHGIIIGGCECEGICFHCECDEFFFFDEEEEEDDDDDCDDCDDDDBC
@HWI-ST911:111:C0N4WACXX:5:1101:2509:2197 1:N:0:TTAGGC CGATC:1+4=B
NATGAGATAAATCAATTGTCTTTAATGAAGTACAGTCTTTGAATAATGAGTTTTGAACTCTTCTGCAACTTTTTGGAAACTTTAAAGTTTGTAATG
+
#4A2<AADHIIIIIIIIIHHIIIIIIIIIIFGIII@GIIFIIIGIIIIEIDHEHIIIHIIIIIIIIIIICHIIIHHEEDFFFFFEEECEEEADDFC
@HWI-ST911:111:C0N4WACXX:5:1101:3746:2179 1:N:0:TTAGGC CCATC:+11+A
NATGTCATCCATCTTTTCCTATCTAAAAAAGAATCAAAAAAGGGATAGTACAGAGGGAAAGTTCAATCCAGAGGACGATGAAACACTGATTGATGG
+
```

HW-ST911	the unique instrument name
111	the run id
C0N4WACXX	the flowcell id
5	flowcell lane
1101	tile number within the flowcell lane
2249	'x'-coordinate of the cluster within the tile
2216	'y'-coordinate of the cluster within the tile
1	the member of a pair, 1 or 2 (paired-end or mate-pair reads only)
Y	Y if the read is filtered, N otherwise
18	0 when none of the control bits are on
TTAGGC, CGATC	index sequence

Alignment before Variant Calling

Precision Mapping

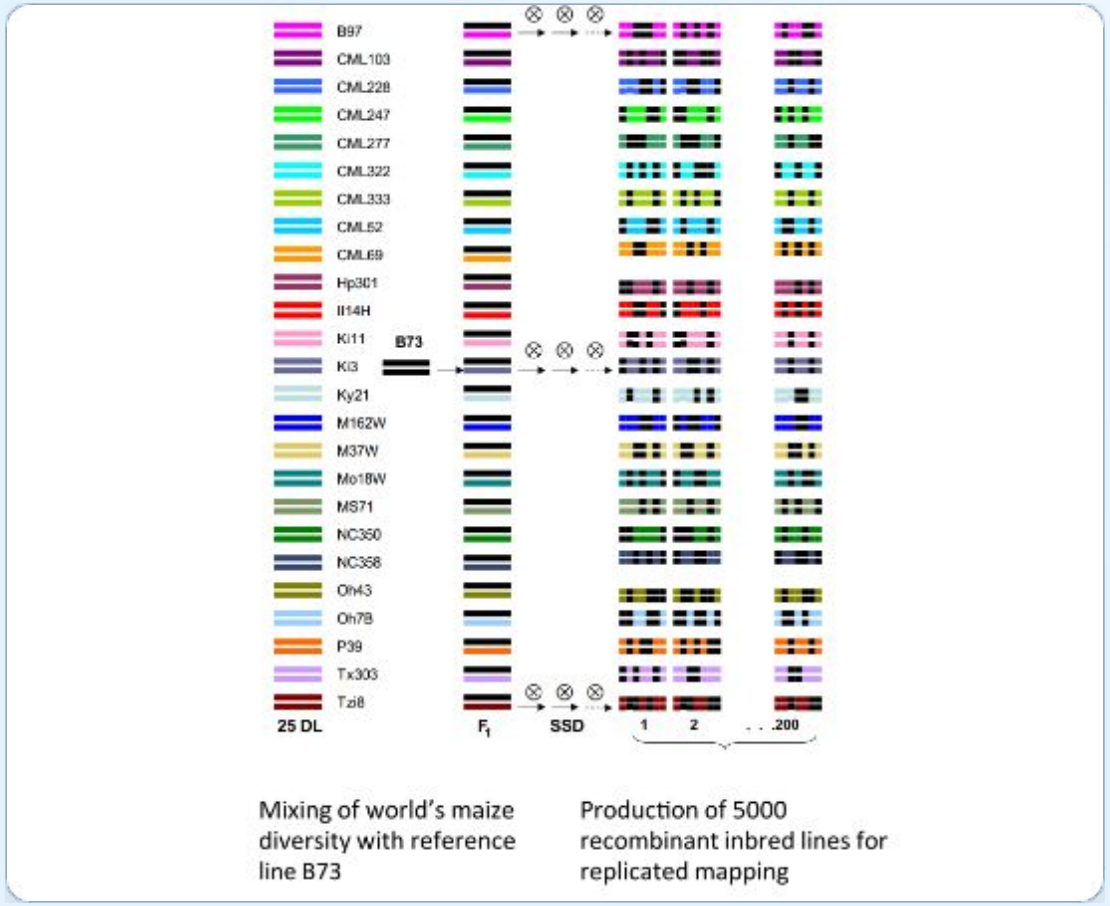
Variant calling depends on how well your reads are mapped to the reference genome. Understanding tools like BWA, Bowtie, or STAR greatly improves your ability to interpret and troubleshoot mutation and SNP calls.



Annotation before Meaning

Functional Context

A list of coordinates has limited value without annotation. Knowing which genes or regulatory regions are affected is what turns raw output into biological insight.



R/Python before Automation

Programming Logic

Effective automation rests on solid programming logic. Learning R or Python for analysis first enables you to write the scripts that later automate your entire project.

Bulk before Single-Cell

Bulk RNA-seq

Learn the foundations of normalization and differential expression across populations.

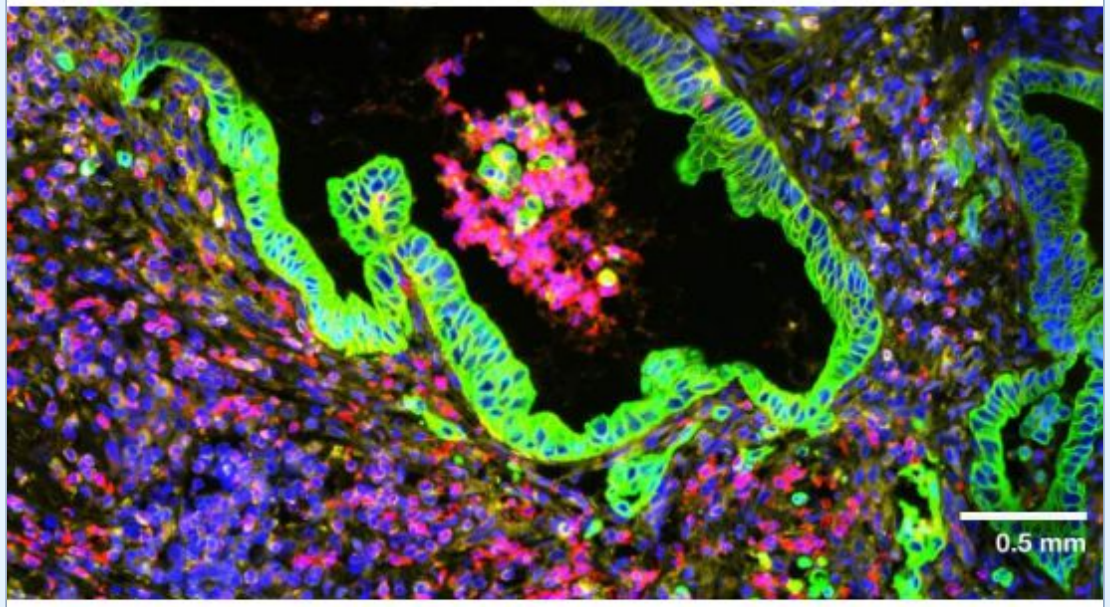
Single-Cell

Then apply those concepts at a more granular level, dealing with higher noise and cell-to-cell variation.

Single-Cell before Spatial

Thinking in Space

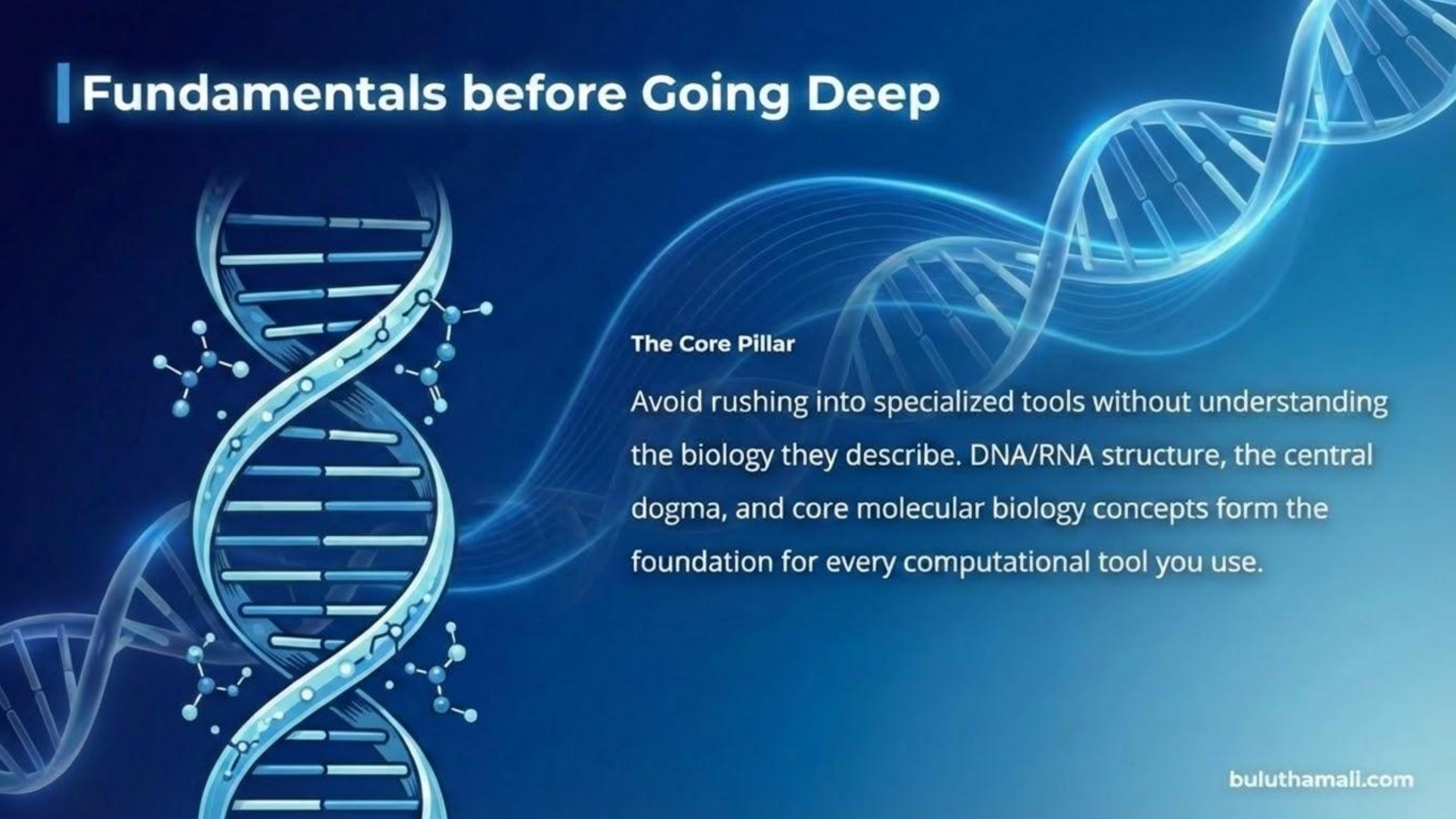
Spatial omics adds a 2D/3D layer to single-cell data. Understanding cell types, clusters, and basic single-cell workflows is a natural step before mapping them back into tissue architecture.



“If it cannot be reproduced by another researcher with the same data, it is not science.”

Build your foundation on Git, containers, and transparent methods so that others can reproduce your work before you make big claims.

Fundamentals before Going Deep



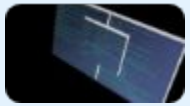
The Core Pillar

Avoid rushing into specialized tools without understanding the biology they describe. DNA/RNA structure, the central dogma, and core molecular biology concepts form the foundation for every computational tool you use.

Not a rule, just a reminder.

Order can matter more than tools. Follow your curiosity, but mind your foundations.

Image Sources



https://www.linuxjournal.com/sites/default/files/styles/360_250/public/nodeimage/story/revolutionizing-command-line-interface-with-dynamic-themes.jpg?itok=eRmxY18T

Source: www.linuxjournal.com

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https://myadlm.org/~/media/project/myadlm/myadlm/images/publications/cln/articles/2020/march/figure-1.jpg

Source: myadlm.org

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https://picture-search.skywork.ai/aippt/image/sheet/a95edc1367ccfae3278eee15f9945e97_zimage.jpg

Source: skywork.ai

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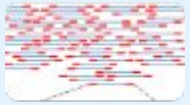
https://learn.gencore.bio.nyu.edu/ngs-file-formats/fastq-format/

Source: learn.gencore.bio.nyu.edu



https://biocorecrg.github.io/PHINDaccess_RNAseq_2020/images/800px-Mapping_Reads.png

Source: biocorecrg.github.io

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https://bioinformaticsworkbook.org/dataAnalysis/GenomeAssembly/GeneticMaps/assets/NAM-fig1.png

Source: bioinformaticsworkbook.org

Image Sources

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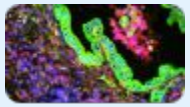
Source: python.plainenglish.io



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